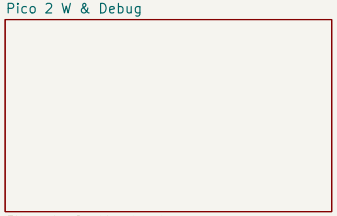
Unijoysticle 2 C64 — Clean-Sheet Pico 2 W Architecture
- MCU: Raspberry Pi Pico 2 W (RP2350 + CYW43439, Dual BR/EDR + BLE, PIO State Machines)
- Power: <45mA from C64 +5V via 250mA PTC Polyfuse + Ferrite Pi-Filter + DMG2305UX Ideal Diode
- Analog Emulation: 5.1k POTX/POTY pull-ups + 6.8k/22k (3.24V) 5V-safe SYNC dividers + 74HCT05

C64 Ports



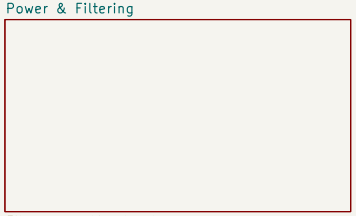
File: c64_ports.kicad_sch

Pico 2 W & Debug



File: pico2w.kicad_sch

Power & Filtering



File: power.kicad_sch

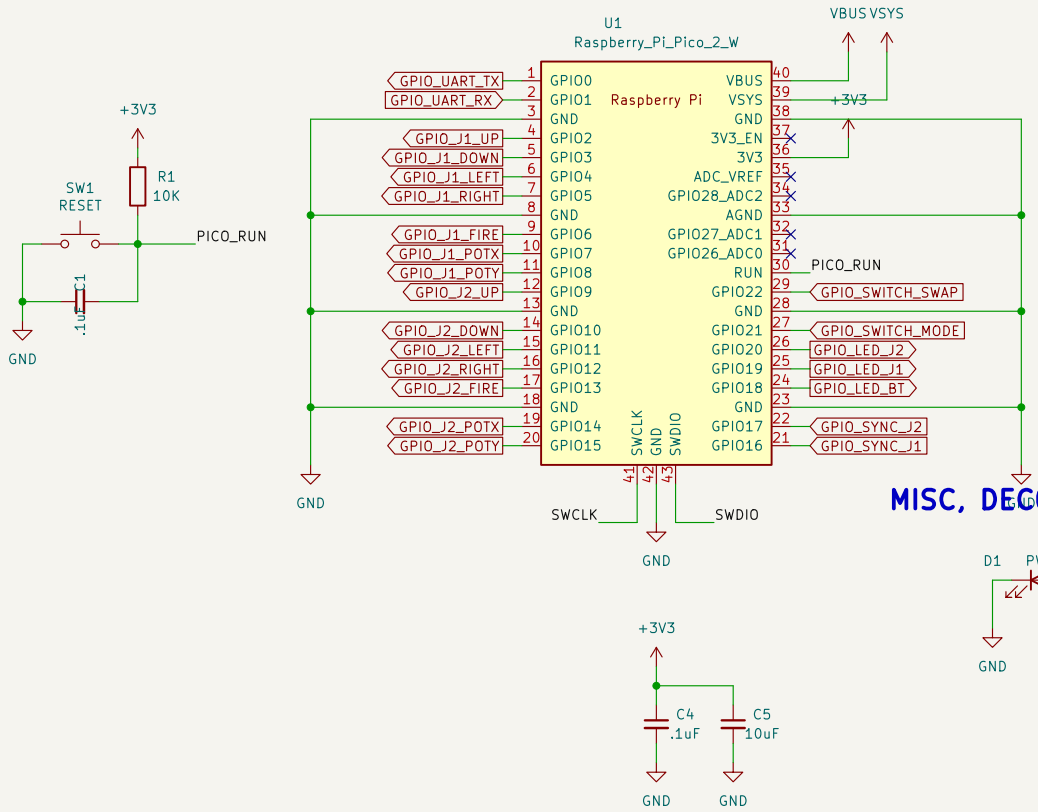
Full C64 Joystick, 1351 Mouse, and 1311/1312 Paddle Emulation
Clean-sheet design with Raspberry Pi Pico 2 W (RP2350 + CYW43439)
**Ricardo Quesada**

Sheet: /
File: unijoysticle2_c64_pico.kicad_sch

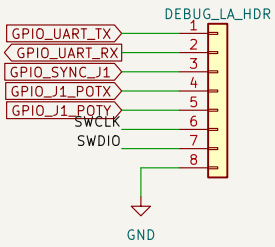
***Title: Unijoysticle 2 C64 (Pico 2 W Edition)***

| Size: A4 | Date: 2026-10-23 | **Rev: D** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 1/4 |

# RASPBERRY PI PICO 2 W (RP2350 + CYW43439)

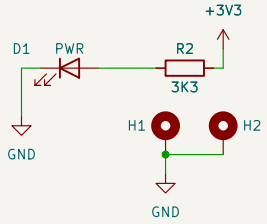



## DEBUG & LOGIC ANALYZER



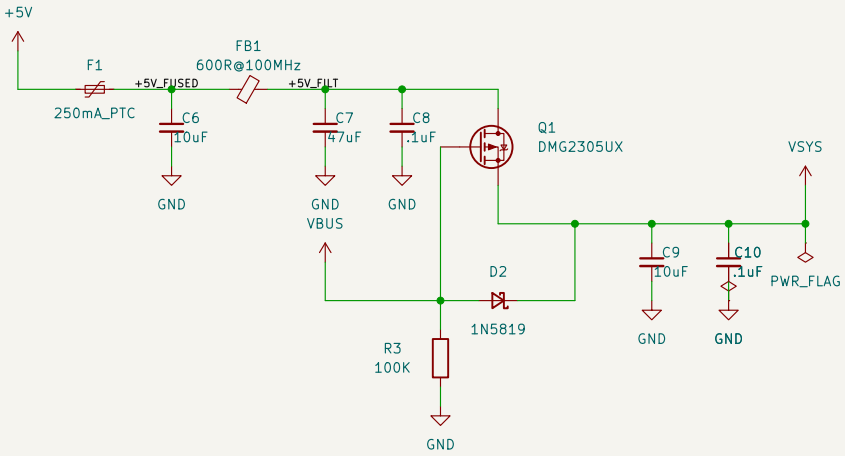

NOTE: GP26–GP28 (ADC0–ADC2) are intentionally left unconnected because RP2350 ADC pads have an internal reverse diode to IOVDD (3.3V) (Pico 2 W Datasheet Section 3.2 Note, p. 13). SYNC_1/2 use 5V fail-safe FT pads GP16/GP17.

## MISC, DECOUPLING & MOUNTING




| | |
|---|---|
| Raspberry Pi Pico 2 W (RP2350 + CYW43439) + 1x8 Debug Header | |
| **Ricardo Quesada** | |
| Sheet: /Pico 2 W & Debug/ | |
| File: pico2w.kicad_sch | |
| ***Title: Unijoysticle 2 C64 (Pico 2 W & Debug)*** | |
| Size: A4 | Date: 2026-10-23 |
| KiCad E.D.A. 10.0.7 | Id: 2/4 |
| | **Rev: D** |

# C64 +5V INPUT PROTECTION & FERRITE PI-FILTER




ZERO-DROP P-FET IDEAL DIODE + USB VBUS OR-ING
(Pico 2 W Datasheet Section 3.4, Figure 10, p. 16)
- When USB is unplugged (VBUS=0V): R3 pulls Q1 Gate to GND, turning Q1 fully ON (Rds_on = 52mOhm, <3mV drop at 45mA).
- When USB is plugged in (VBUS=5V): Q1 Gate is pulled to 5V, turning Q1 OFF to block back-feeding into C64 +5V rail.

250mA PTC Polyfuse + Ferrite Pi-Filter + DMG2305UX Ideal Diode OR-ing

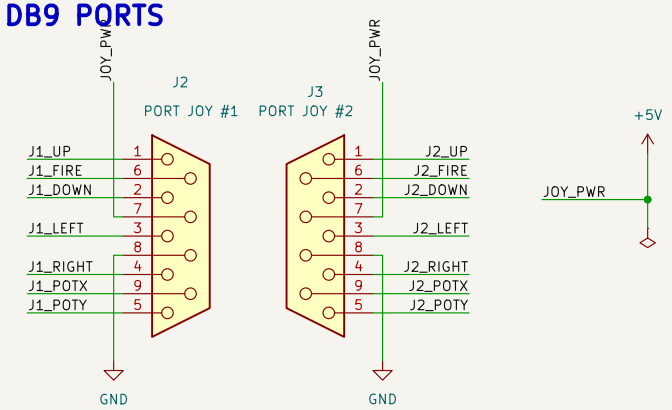
**Ricardo Quesada**

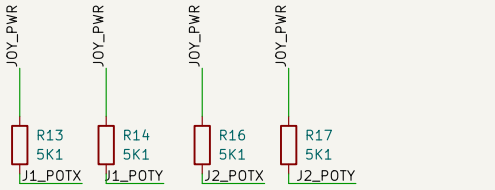
Sheet: /Power & Filtering/
File: power.kicad_sch

***Title: Unijoysticle 2 C64 (Power & Filtering)***

| Size: A4 | Date: 2026-10-23 | **Rev: D** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 3/4 |

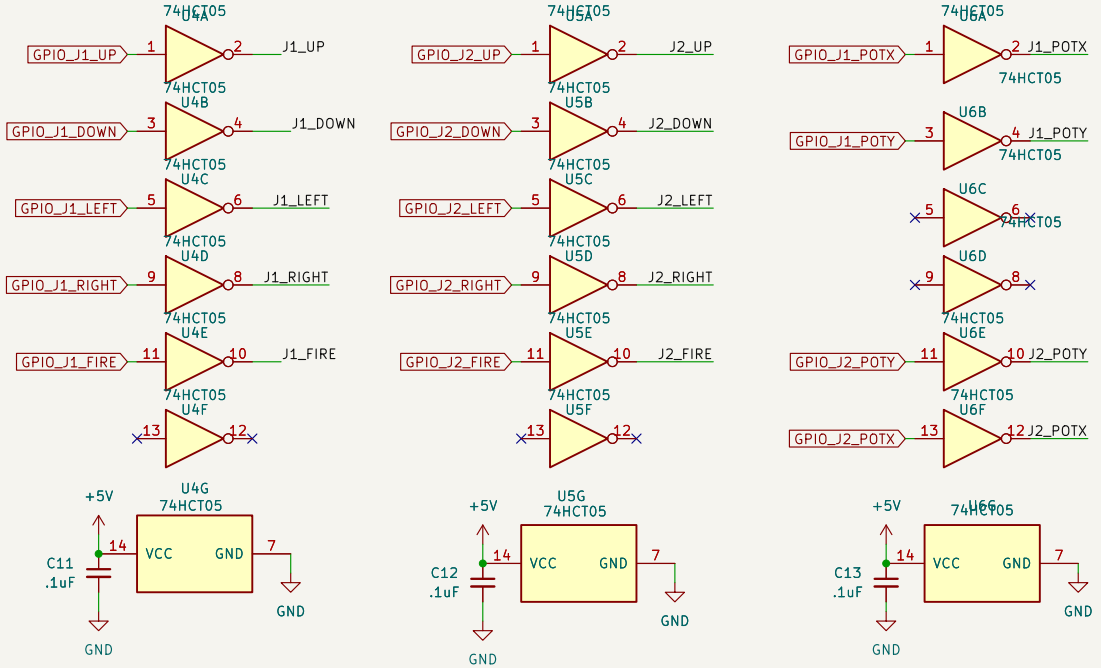
## DB9 PORTS



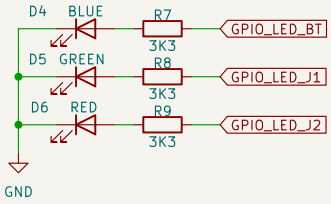

5.1k POTX/POTY Pull-Ups (1351 Mouse & 1311/1312 Paddles)

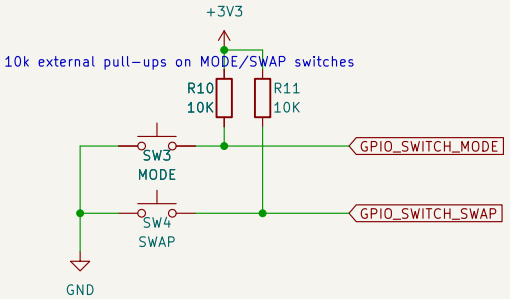
## TTL OPEN-DRAIN BUFFERS (74HCT05)




74HCT05 (V_IH=2.0V max) powered from C64 +5V (VDD)
guarantees full 630mV TTL noise margin with 3.3V RP2350 GPIOs.

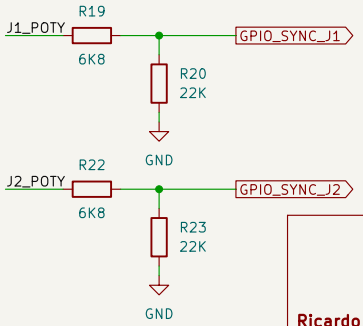
## LEDS & SWITCHES




## 5V-SAFE SID SYNC DIVIDERS (6.8k / 22k -> 3.24V MAX)

6.8k series + 22k pull-down divides C64 5V POTY
to 3.24V max (<3.6V absolute max), zero ESD diode injection.





Ricardo Quesada


Sheet: /C64 Ports/
File: c64_ports.kicad_sch

**Title: Unijoysticle 2 C64 (DB9 Ports, 74HCT05 & SID SYNC)**

| Size: A4 | Date: 2026-10-23 | Rev: D |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 4/4 |